

Math 118 – Fourier Analysis

Chapter-by-chapter notes on the important ideas

Bogges & Narcowich, *A First Course in Wavelets with Fourier Analysis* (2nd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Taking notes

Find the chapter banner, then type under the section heading as you read:

`\notesec{1A}{Span and Linear Independence}`

Record big ideas with `idea`, definitions with `definition`, theorems with `theorem`, `lemma`, `proposition`, examples with `example`, and warnings with `remark` — all numbered within the chapter. Add a missing section by copying a `\notesec` slot; add a chapter with `\chapterbanner`.

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Chapter 0 Inner Product Spaces

Bogges & Narcowich: inner products, L^2 and ℓ^2 , orthogonality, projections, Gram–Schmidt, adjoints, least squares

§0.1 Motivation

§0.2 Definition of Inner Product

§0.3 The Spaces L^2 and ℓ^2

§0.4 Schwarz and Triangle Inequalities

§0.5 Orthogonality

§0.6 Linear Operators and Their Adjoints

§0.7 Least Squares and Linear Predictive Coding

Chapter 1 Fourier Series

Bogges & Narcowich: computing Fourier series, convergence theorems at and between points

§1.1 Introduction

§1.2 Computation of Fourier Series

§1.3 Convergence Theorems for Fourier Series

Chapter 2 The Fourier Transform

Bogges & Narcowich: inversion, basic properties, convolution, linear filters, the sampling theorem, the uncertainty principle

§2.1 Informal Development of the Fourier Transform

§2.2 Properties of the Fourier Transform

§2.3 Linear Filters

§2.4 The Sampling Theorem

§2.5 The Uncertainty Principle

Chapter 3 Discrete Fourier Analysis

Bogges & Narcowich: the DFT and FFT, discrete signals and filters, the z -transform

§3.1 The Discrete Fourier Transform

§3.2 Discrete Signals

§3.3 Discrete Signals & Matlab

Chapter 4 Haar Wavelet Analysis

Boggess & Narcowich: the Haar scaling function and wavelet, decomposition and reconstruction algorithms

§4.1 Why Wavelets?

§4.2 Haar Wavelets

§4.3 Haar Decomposition and Reconstruction Algorithms

§4.4 Summary

Chapter 5 Multiresolution Analysis

Boggess & Narcowich: the multiresolution framework, decomposition/reconstruction algorithms, Fourier transform criteria

§5.1 The Multiresolution Framework

§5.2 Implementing Decomposition and Reconstruction

§5.3 Fourier Transform Criteria

Chapter 6 The Daubechies Wavelets

Bogges & Narcowich: Daubechies' construction, moments and smoothness, computational issues, dyadic values

§6.1 Daubechies' Construction

§6.2 Classification, Moments, and Smoothness

§6.3 Computational Issues

§6.4 The Scaling Function at Dyadic Points

Chapter 7 Other Wavelet Topics

Bogges & Narcowich: computational complexity, higher dimensions, decomposition vs. reconstruction, the wavelet transform

§7.1 Computational Complexity

§7.2 Wavelets in Higher Dimensions

§7.3 Relating Decomposition and Reconstruction

§7.4 The Wavelet Transform
